

GammaLoop

Differential cross-sections with Local Unitarity

1 Overview

GammaLoop computes differential collider cross-sections with the Local Unitarity method. It is an application and research codebase rather than a thin numerical library: process generation, model import, integrand construction, integration, observables, persistence, and diagnostics meet in one stateful workflow.

Primary interface

The command-line interface is the primary user interface. A run card creates a state, executes commands, and leaves a reusable working directory. The Rust and Python APIs load the same persisted state and expose selected structured operations; they do not define a second, independent execution model.

1.1 A stateful run

A typical source checkout is built and exercised from the repository root:

```
just build-cli-release
./gammaLoop ./examples/cli/gg_hhh/1L/gg_hhh_1L.toml
```

The run card imports a model, generates the requested process, executes its command blocks, and records the resulting state. Resume that directory explicitly for subsequent work:

```
./gammaLoop -s ./examples/cli/gg_hhh/1L/state \
  run integrate_physical -c "quit -o"
```

The persisted run.toml records how to replay the run. The settings files describe global and default runtime configuration, while processes/ contains process-specific state. Ordinary runs create gammaLoop_state/ unless a different state path is supplied.

1.2 Installation and external tools

GammaLoop is currently developed primarily from source. The supported development path is the repository's Nix shell, or a local Rust toolchain together with just, a recent GNU toolchain, and Python 3.11 or newer when building bindings. FORM 4.2.1 or newer is needed for analytical integration of integrated UV counterterms. UFO model import also needs the Python ufo-model-loader package.

```
nix develop
just build-cli
just build-api
```

Diagram rendering is a separate concern and uses Clinnet and Typst. Building the CLI does not imply that these drawing tools are installed.

1.3 Related crates

How the crates fit together

Linnet provides the half-edge graph model and graph algorithms. Spenso provides typed tensors, tensor structures, and network execution. Idenso handles symbolic tensor identities and algebraic simplification. Vakint matches and evaluates vacuum-integral topologies. GammaLoop combines these components into a collider-calculation workflow.

For a deeper view of how the command layer, state, integrands, and external components fit together, see the [architecture overview](#).

1.4 Where to begin

- Use built-in `--help` for the CLI options supported by your installed version.
- Start with the included `gg_hhh` run card for a complete state lifecycle.
- Use the Rust or Python API only when a program needs structured access to loaded state or sample-evaluation results.
- Check prerequisites before starting an expensive integration; some workflows require a Symbolica license or external tools such as FORM.

2 Tutorial

This tutorial builds the GammaLoop command-line interface and uses the included one-loop `g g -> h h h` run card to create a state that can be inspected and resumed. Compilation and process generation can take longer than a `--help` smoke test.

2.1 Prerequisites

Work from a GammaLoop source checkout. The supported development environment is the repository Nix shell; without Nix you need Rust 1.85 or newer, just, a recent GNU toolchain, and the dependencies described in the root README. The bundled Standard Model used below does not need the UFO loader. A Symbolica license may be required by your build and use case.

```
nix develop
just build-cli
./gammaLoop --help
```

The last command should print the one-shot CLI options and command tree. The `./gammaLoop` wrapper selects the binary built under `target/dev-optimize/`.

2.2 Generate an example process

The run card declares a state folder, settings, and named command blocks. Run its `generate` block and persist the result on exit:

```
./gammaLoop --clean-state \
  ./examples/cli/gg_hhh/1L/gg_hhh_1L.toml \
  run generate -c "quit -o"
```

`--clean-state` removes the resolved state first`

Use it only for this first, reproducible run. Omit it when resuming work you want to keep. The example card resolves its state to `examples/cli/gg_hhh/1L/state` and requests ten worker cores; make a private copy of the card before changing either value.

The block imports `sm-default.json`, generates the selected one-loop pentagon contribution, builds its integrand, saves DOT data, and writes the state. A successful run exits without an error and leaves at least `run.toml`, `global_settings.toml`, `default_runtime_settings.toml`, and `processes/` in the state folder. `run.toml` records the commands and settings needed to replay the run; it is not merely a log file.

2.3 Resume the saved state

Load the saved state and ask the active session to display its processes:

```
./gammaloop -s ./examples/cli/gg_hhh/1L/state \  
run -c "display processes; quit -o"
```

Success means the display includes the generated `gg_hhh` process and the command exits while keeping the same state. You can now run the card's `integrate_euclidean` or `integrate_physical` block, but those are deliberately beyond the first-success path because they request a substantial Monte Carlo integration.

2.4 Troubleshooting and next steps

- If the wrapper cannot find a binary, rerun `just build-cli` and confirm that `target/dev-optim/gammaloop` exists.
- If compilation or process generation exhausts local resources, copy the card and reduce the values under `cli_settings.global.n_cores`; do not edit a state midway through generation.
- If startup reports a state fingerprint or settings mismatch, replay the card with a new state folder instead of transplanting only its `processes/` directory.
- Use `./gammaloop help generate` to inspect the flags supported by your installed version, then continue with the process-generation manual and the CLI reference.

Example run card

See the complete `gg_hhh` run card before adapting the process, state location, or resource settings for your own calculation.

3 Process generation and state workflows

GammaLoop describes a calculation with a TOML run card and stores its reusable state in a directory. Start the command-line interface through the repository wrapper (`./gammaloop`) or the Cargo-built executable, then generate processes, integrands, and integration jobs within that state.

3.1 Choose a generation mode

The interactive command tree separates cross sections, amplitudes, and operations on processes already stored in the active state:

```
generate xs INITIAL > FINAL ...  
generate amp INITIAL > FINAL ...  
generate existing ...  
generate help xs
```

`xs` constructs forward-scattering graphs and filters compatible Cutkosky cuts. `amp` constructs ordinary amplitudes. Their final-state lists therefore have different meanings even when the printed particle syntax is identical. Use `generate help <mode>` inside the stateful CLI to see the flags supported by your installed version.

3.2 Process specification grammar

Process specifications accept `to` or `>` between space-separated initial and final states. Braces express alternatives in a cross-section final state, while `{}` denotes an empty state:

```
e+ e- > mu+ mu-
e+ e- > { Z Z, a a, H H }
{} > {}
```

A slash vetoes particles and a vertical bar selects an allow-list. The words `veto` and `only` are equivalent forms. Amplitude coupling constraints use the coupling name directly; powered constraints apply to the complete cross section:

```
e+ e- > d d~ g / u c QED==2 QCD>=2 QCD<=4
e+ e- > mu+ mu- | g ghG QED^2==2 QCD^2>=2
```

The bracketed perturbative block is intentionally distinct from ordinary flags. `{n}` fixes the amplitude loop count (or the sum across the two cut sides), `{{n}}` fixes the forward-graph loop count, and `QCD=n` or the shorthand `QCD` selects a relative perturbative order:

```
e+ e- > Z [ {1} {{2}} QCD=2 QED=1 ]
e+ e- > Z [ QCD ]
```

Resolve names through the active model

Particle names, antiparticles, coupling-order names, and allowed interactions come from the model loaded in the current state. A syntactically valid specification can still be rejected when that model does not define a requested name.

3.3 Filters, grouping, and multiplicity

Generation flags control topology filters, loop ranges, cut blobs and spectators, symmetrization, and numerator-aware grouping. These choices are physics-significant. In particular, grouping can combine graph multiplicities and numerator ratios; external-fermion symmetrization requires the caller to retain the associated ordering signs. Run `generate help <mode>` before choosing filters to see the available aliases, defaults, values, and argument counts.

The generated graph's overall factor keeps separate contributions for automorphisms, internal fermion loops, external-fermion ordering, numerator-independent symmetry grouping, and numerator-dependent grouping. Persist the state and inspect that factor when auditing diagram normalization instead of replacing it with an assumed factorial.

3.4 From generation to evaluation

A reproducible run keeps its stages explicit:

- `import` or select the model before resolving a process specification;
- `generate` the process and inspect the retained diagrams;
- `generate` the requested integrand representation;
- set kinematics, sampling, stability, and subtraction settings;
- evaluate samples or run an integration command block;
- exit with persistence enabled when the state should be resumed.

Keep the complete state together

`run.toml` records boot settings, reusable command blocks, and top-level commands. Generated process and

integrand artifacts live beside it. Resume with `-s STATE_FOLDER`; replay the run card to reconstruct the state from scratch. Copy or archive the entire state directory rather than moving a generated subdirectory on its own, because settings and fingerprints are checked together.

4 CLI, Rust, and Python APIs

4.1 Rust packages

The Rust API spans two primary packages.

- `gammalooprs` contains the physics implementation, data model, integrands, integration, observables, and lower-level algorithms.
- `gammaloop-api` provides the supported state-loading API, session/command integration, CLI assembly, and the PyO3 extension used by the Python distribution.

Load a state with `StateLoadOption::load()`. Its options select a state directory, boot card, logging overrides, read-only behavior, and optional settings overrides. Once loaded, callers can create a CLI-style session or use dedicated structured operations. A raw command string is the compatibility fallback, not a replacement for a typed operation where one exists.

```
use std::path::PathBuf;
use gammaloop_api::{state::CommandHistory, StateLoadOption};

let mut loaded = StateLoadOption {
    state_folder: Some(PathBuf::from("./state")),
    ..StateLoadOption::default()
}.load()?;

let command = CommandHistory::from_raw_string("display settings global"?);
loaded.cli_session().execute_command(command)?;
```

Some Rust types are public so that GammaLoop's packages can work together, but are not intended as stable integration points. Prefer the state-loading and structured-operation APIs described here; start with the [curated Rust API](#) and use its [Rustdoc](#) links when you need lower-level types or trait implementations.

4.2 Python packaging

A standalone GammaLoop distribution

The Python distribution and import package are both named `gammaloop`. Its compiled backend is `gammaloop._gammaloop`, which applications normally access through the public package. This is separate from the `symbolica.community.*` modules used by `Spenso`, `Idenso`, and `Vakint`; installing GammaLoop does not install those community modules as independent packages.

The main Python entry point is `GammaLoopAPI`:

```
from gammaloop import GammaLoopAPI

gl = GammaLoopAPI(
    state_folder="./state",
    boot_commands_path="./run.toml",
)
gl.run("display settings global")
```

The Python package requires Python 3.11 or newer. Run `just build-api` when building the bindings from a source checkout.

4.3 Sample evaluation and precision

`evaluate_sample` returns one sample result and the observable bundle for that one-sample batch. `evaluate_samples` returns per-sample results plus a batch-global observable bundle. Both the ordinary Rust and Python endpoints expose numeric results through an `f64` contract, even if arbitrary precision was used internally. Rust-only callers that must retain the active precision use `evaluate_sample_precise` and `evaluate_samples_precise`.

4.4 CLI and settings reference

Use `./gamma-loop --help` and the generated CLI and settings reference for command names, aliases, flags, positional arguments, defaults, possible values, and setting paths. The tutorials explain how commands and settings combine into a persistent workflow.

5 Releases and change history

GammaLoop versions are identified by repository tags together with the version of the Python distribution and the `gamma-loop-rs` Rust package. These components can evolve independently, so record all of them when reporting a result or asking for support.

5.1 Choosing the matching documentation

The `latest` channel follows current development. Documentation under `snapshots/<tag>` stays with a tagged release. Use the snapshot matching your checkout whenever command options, state files, or API signatures differ from `latest`.

5.2 Before upgrading

Release notes currently live with repository tags and their linked pull requests rather than in one consolidated changelog. Review them before upgrading, especially when you depend on persisted states, CLI scripts, Python bindings, or numerical results. Keep a copy of valuable state directories and re-run validation samples after changing versions.

Information to include in a report

Include the repository tag or commit, the Python distribution version, the `gamma-loop-rs` version, the run card, and any relevant settings overrides. This makes a calculation easier to reproduce and distinguishes a software change from a configuration change.

API reference

The packages available in this version are listed below. Follow an item link for its complete language reference.

gammaloops

GammaLoop's supported Rust entry points. The complete compiled surface remains available in the Rustdoc sidecar.

GammaLoopContext

Combines the state-map and model capabilities required by GammaLoop computations.

Combines the state-map and model capabilities required by GammaLoop computations.

```
pub trait GammaLoopContext : HasStateMap + HasModel {}
```

Supported usage

```
fn accepts_context<C: gammaloops::GammaLoopContext>(_context: &C) {}
```

Exhaustive reference · Source

HasModel

Provides shared access to the physics model owned by a computation context.

Provides shared access to the physics model owned by a computation context.

```
pub trait HasModel { fn get_model(& self) -> & Model; }
```

Supported usage

```
fn model_owner<C: gammaloops::HasModel>(_context: &C) {}
```

get_model

Kind: Method

```
fn get_model(& self) -> & Model;
```

Exhaustive reference · Source

Install interrupt handling

Installs GammaLoop's process interrupt handler before a long-running integration.

Installs GammaLoop's process interrupt handler before a long-running integration.

```
fn set_interrupt_handler()
```

Supported usage

```
gammaloops::set_interrupt_handler();
```

Exhaustive reference · Source

gammaloop-api

GammaLoop's stable facade and command API. Use this curated boundary for applications while consulting the full Rustdoc for exhaustive details.

State loading options

Selects the state folder, boot commands, settings overrides, logging, and read-only mode.

Selects the state folder, boot commands, settings overrides, logging, and read-only mode.

```
#[derive(Debug, Clone, PartialEq, Default)] pub struct StateLoadOption
{
    pub clean_state : bool, pub boot_commands_path : Option < PathBuf > , pub
    state_folder : Option < PathBuf > , pub model_file : Option < PathBuf > ,
    pub trace_logs_filename : Option < String > , pub level : Option <
    LogLevel > , pub logfile_level : Option < LogLevel > , pub logging_prefix
    : Option < LogFormat > , pub read_only_state : bool, pub
    settings_global_path : Option < PathBuf > , pub
    settings_runtime_defaults_path : Option < PathBuf > ,
}
```

Reference feature matrix: cli (item is available without an additional gate)

Supported usage

```
let options = gammaloop_api::StateLoadOption::default();
```

clean_state

Kind: Field

bool

boot_commands_path

Kind: Field

Option < PathBuf >

state_folder

Kind: Field

Option < PathBuf >

model_file

Kind: Field

Option < PathBuf >

trace_logs_filename

Kind: Field

Option < String >

level

Kind: Field

Option < LogLevel >

logfile_level

Kind: Field

Option < LogLevel >

logging_prefix

Kind: Field

Option < LogFormat >

read_only_state

Kind: Field

bool

settings_global_path

Kind: Field

Option < PathBuf >

settings_runtime_defaults_path

Kind: Field

Option < PathBuf >

Exhaustive reference · Source

Load a GammaLoop state

Loads the selected state and returns the supported session boundary.

Loads the selected state and returns the supported session boundary.

```
fn load(self) -> Result < LoadedState >
```

Reference feature matrix: cli (item is available without an additional gate)

Parameter	Type
self	self

Returns: Result < LoadedState >

Supported usage

```
let loaded = gammaloop_api::StateLoadOption::default().load()?;
```

Exhaustive reference · Source

Create a CLI session

Borrows a loaded state as a command session while keeping state ownership explicit.

Borrows a loaded state as a command session while keeping state ownership explicit.

```
fn cli_session(& mut self) -> CliSession < '_ >
```

Reference feature matrix: cli (item is available without an additional gate)

Parameter	Type
self	& mut self

Returns: CliSession < '_ >

Supported usage

```
let mut loaded = gammaloop_api::StateLoadOption::default().load()?; let session = loaded.cli_session();
```

Exhaustive reference · Source

Command-line settings

Owns the generated command-line, global, session, and state-loading settings.

Owns the generated command-line, global, session, and state-loading settings.

```
#[derive(Debug, Clone, Deserialize, Serialize, JsonSchema)]
#[serde(default, deny_unknown_fields)] pub struct CLISettings
{
    #[serde(skip_serializing_if = "is_true")] pub try_strings : bool,
    #[serde(skip_serializing_if = "is_false")] pub override_state : bool,
```

```

    #[serde(skip_serializing_if = "IsDefault::is_default")] pub state :
    StateSettings, #[serde(skip_serializing_if = "IsDefault::is_default")] pub
    global : GlobalSettings, #[serde(skip)] #[schemars(skip)] pub session :
    SessionSettings,
}

```

Reference feature matrix: cli (item is available without an additional gate)

Supported usage

```
let settings = gammaloop_api::CLISettings::default();
```

try_strings

Kind: Field

bool

override_state

Kind: Field

bool

state

Kind: Field

StateSettings

global

Kind: Field

GlobalSettings

session

Kind: Field

SessionSettings

Exhaustive reference · Source

GammaLoop command

Runs the Clap-derived command tree for state creation, inspection, generation, and evaluation.

Runs the Clap-derived command tree for state creation, inspection, generation, and evaluation.

```
gammaloop [OPTIONS] [COMMAND]
```

Reference feature matrix: cli (item is available without an additional gate)

Supported usage

```
gammaloop --help
```

try_strings

Kind: Field

bool

override_state

Kind: Field

bool

state

Kind: Field

StateSettings

global

Kind: Field

GlobalSettings

session

Kind: Field

SessionSettings

Exhaustive reference · Source

gammaLoop-python

Exports registered by gammaLoop.

GammaLoopAPI

Stateful entry point for loading, evaluating, and integrating GammaLoop processes.

Stateful entry point for loading, evaluating, and integrating GammaLoop processes.

class GammaLoopAPI:

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```
from gammaLoop._gammaLoop import GammaLoopAPI
help(GammaLoopAPI)
```

__new__

Kind: AssociatedFunction

```
def __new__(cls, state_folder: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, boot_commands_path: typing.Optional[builtins.str | os.PathLike
| pathlib.Path] = None, model_file: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, trace_logs_filename: typing.Optional[builtins.str] = None,
level: typing.Optional[LogLevel] = None, logfile_level: typing.Optional[LogLevel] =
None, logging_prefix: builtins.object | None = None, read_only_state: builtins.bool
= ..., settings_global_path: typing.Optional[builtins.str | os.PathLike |
pathlib.Path] = None, settings_runtime_defaults_path: typing.Optional[builtins.str |
os.PathLike | pathlib.Path] = None, clean_state: builtins.bool = ...) ->
GammaLoopAPI:
```

state_folder

Kind: Parameter

state_folder: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None

Default: None

boot_commands_path

Kind: Parameter

boot_commands_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None

Default: None

model_file**Kind:** Parameter`model_file: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**trace_logs_filename****Kind:** Parameter`trace_logs_filename: typing.Optional[builtins.str] = None`**Default:** None**level****Kind:** Parameter`level: typing.Optional[LogLevel] = None`**Default:** None**logfile_level****Kind:** Parameter`logfile_level: typing.Optional[LogLevel] = None`**Default:** None**logging_prefix****Kind:** Parameter`logging_prefix: builtins.object | None = None`**Default:** None**read_only_state****Kind:** Parameter`read_only_state: builtins.bool = ...`**Default:** ...**settings_global_path****Kind:** Parameter`settings_global_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**settings_runtime_defaults_path****Kind:** Parameter`settings_runtime_defaults_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**clean_state****Kind:** Parameter`clean_state: builtins.bool = ...`

Default: ...

evaluate_sample

Kind: Method

```
def evaluate_sample(self, point: typing.Sequence[builtins.float], process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None, use_arb_prec: builtins.bool = ..., minimal_output: builtins.bool = ...,
return_events: typing.Optional[builtins.bool] = None, momentum_space: builtins.bool
= ..., integrator_weight: typing.Optional[builtins.float] = None, discrete_dim:
typing.Optional[typing.Sequence[builtins.int]] = None, graph_name:
typing.Optional[builtins.str] = None, orientation: typing.Optional[builtins.int] =
None) -> EvaluationResult:
```

point

Kind: Parameter

point: typing.Sequence[builtins.float]

process_id

Kind: Parameter

process_id: typing.Optional[builtins.int] = None

Default: None

integrand_name

Kind: Parameter

integrand_name: typing.Optional[builtins.str] = None

Default: None

use_arb_prec

Kind: Parameter

use_arb_prec: builtins.bool = ...

Default: ...

minimal_output

Kind: Parameter

minimal_output: builtins.bool = ...

Default: ...

return_events

Kind: Parameter

return_events: typing.Optional[builtins.bool] = None

Default: None

momentum_space

Kind: Parameter

momentum_space: builtins.bool = ...

Default: ...

integrator_weight**Kind:** Parameter

integrator_weight: typing.Optional[builtins.float] = None

Default: None**discrete_dim****Kind:** Parameter

discrete_dim: typing.Optional[typing.Sequence[builtins.int]] = None

Default: None**graph_name****Kind:** Parameter

graph_name: typing.Optional[builtins.str] = None

Default: None**orientation****Kind:** Parameter

orientation: typing.Optional[builtins.int] = None

Default: None**evaluate_samples****Kind:** Method

```
def evaluate_samples(self, points: numpy.typing.NDArray[numpy.float64], process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None, use_arb_prec: builtins.bool = ..., minimal_output: builtins.bool = ...,
return_events: typing.Optional[builtins.bool] = None, momentum_space: builtins.bool
= ..., integrator_weights: typing.Optional[numpy.typing.NDArray[numpy.float64]] =
None, discrete_dims: numpy.typing.NDArray[numpy.unsignedinteger] | None = None,
graph_names: typing.Optional[typing.Sequence[typing.Optional[builtins.str]]] = None,
orientations: typing.Optional[typing.Sequence[typing.Optional[builtins.int]]] = None)
-> BatchEvaluationResult:
```

points**Kind:** Parameter

points: numpy.typing.NDArray[numpy.float64]

process_id**Kind:** Parameter

process_id: typing.Optional[builtins.int] = None

Default: None**integrand_name****Kind:** Parameter

integrand_name: typing.Optional[builtins.str] = None

Default: None

use_arb_prec**Kind:** Parameter

use_arb_prec: builtins.bool = ...

Default: ...**minimal_output****Kind:** Parameter

minimal_output: builtins.bool = ...

Default: ...**return_events****Kind:** Parameter

return_events: typing.Optional[builtins.bool] = None

Default: None**momentum_space****Kind:** Parameter

momentum_space: builtins.bool = ...

Default: ...**integrator_weights****Kind:** Parameter

integrator_weights: typing.Optional[numpy.typing.NDArray[numpy.float64]] = None

Default: None**discrete_dims****Kind:** Parameter

discrete_dims: numpy.typing.NDArray[numpy.unsignedinteger] | None = None

Default: None**graph_names****Kind:** Parameter

graph_names: typing.Optional[typing.Sequence[typing.Optional[builtins.str]]] = None

Default: None**orientations****Kind:** Parameter

orientations: typing.Optional[typing.Sequence[typing.Optional[builtins.int]]] = None

Default: None**import_graphs****Kind:** Method

```
def import_graphs(self, graphs: builtins.str, process_name:
typing.Optional[builtins.str] = None, process_id: typing.Optional[builtins.int] =
```

```
None, integrand_name: typing.Optional[builtins.str] = None, format: builtins.str
= ..., overwrite: builtins.bool = ..., append: builtins.bool = ...) -> None:
```

graphs

Kind: Parameter

graphs: builtins.str

process_name

Kind: Parameter

process_name: typing.Optional[builtins.str] = None

Default: None

process_id

Kind: Parameter

process_id: typing.Optional[builtins.int] = None

Default: None

integrand_name

Kind: Parameter

integrand_name: typing.Optional[builtins.str] = None

Default: None

format

Kind: Parameter

format: builtins.str = ...

Default: ...

overwrite

Kind: Parameter

overwrite: builtins.bool = ...

Default: ...

append

Kind: Parameter

append: builtins.bool = ...

Default: ...

get_lmbs

Kind: Method

```
def get_lmbs(self, graphs: builtins.str, format: builtins.str = ...) ->
builtins.list[builtins.list[tuple[builtins.list[builtins.int],
builtins.list[builtins.int], builtins.dict[builtins.int,
tuple[builtins.list[builtins.int], builtins.list[builtins.int]]]]]:
```

graphs

Kind: Parameter

graphs: builtins.str

format**Kind:** Parameter

format: builtins.str = ...

Default: ...**get_orientations****Kind:** Method

```
def get_orientations(self, graph_name: builtins.str, process_id:
typing.Optional[builtins.int] = None, integrand_name: typing.Optional[builtins.str] =
None) -> builtins.list[builtins.dict[builtins.int, builtins.int]]:
```

graph_name**Kind:** Parameter

graph_name: builtins.str

process_id**Kind:** Parameter

process_id: typing.Optional[builtins.int] = None

Default: None**integrand_name****Kind:** Parameter

integrand_name: typing.Optional[builtins.str] = None

Default: None**get_model****Kind:** Method

```
def get_model(self) -> builtins.str:
```

evaluate**Kind:** Method

```
def evaluate(self, process_id: typing.Optional[builtins.int] = None,
graphs_group_name: typing.Optional[builtins.str] = None, result_path:
typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None, numerical:
builtins.bool = ..., number_of_terms_in_epsilon_expansion:
typing.Optional[builtins.int] = None) -> builtins.str:
```

process_id**Kind:** Parameter

process_id: typing.Optional[builtins.int] = None

Default: None**graphs_group_name****Kind:** Parameter

graphs_group_name: typing.Optional[builtins.str] = None

Default: None

result_path**Kind:** Parameter`result_path: typing.Optional[builtins.str | os.PathLike | pathlib.Path] = None`**Default:** None**numerical****Kind:** Parameter`numerical: builtins.bool = ...`**Default:** ...**number_of_terms_in_epsilon_expansion****Kind:** Parameter`number_of_terms_in_epsilon_expansion: typing.Optional[builtins.int] = None`**Default:** None**import_model****Kind:** Method

```
def import_model(self, model_specifier: builtins.str | os.PathLike | pathlib.Path,
simplify_model: builtins.bool = ...) -> None:
```

model_specifier**Kind:** Parameter`model_specifier: builtins.str | os.PathLike | pathlib.Path`**simplify_model****Kind:** Parameter`simplify_model: builtins.bool = ...`**Default:** ...**list_outputs****Kind:** Method

```
def list_outputs(self) -> tuple[builtins.dict[builtins.str, builtins.int],
builtins.dict[builtins.str, builtins.int]]:
```

get_integrand_info**Kind:** Method

```
def get_integrand_info(self, process_id: typing.Optional[builtins.int] = None,
integrand_name: typing.Optional[builtins.str] = None) -> IntegrandInfo:
```

process_id**Kind:** Parameter`process_id: typing.Optional[builtins.int] = None`**Default:** None**integrand_name****Kind:** Parameter

integrand_name: typing.Optional[builtins.str] = None

Default: None

get_integrand_settings

Kind: Method

```
def get_integrand_settings(self, process_id: typing.Optional[builtins.int] = None,
integrand_name: typing.Optional[builtins.str] = None) -> SettingsValue:
```

process_id

Kind: Parameter

process_id: typing.Optional[builtins.int] = None

Default: None

integrand_name

Kind: Parameter

integrand_name: typing.Optional[builtins.str] = None

Default: None

get_run_history

Kind: Method

```
def get_run_history(self) -> builtins.str:
```

get_global_settings

Kind: Method

```
def get_global_settings(self) -> builtins.str:
```

get_active_command_blocks

Kind: Method

```
def get_active_command_blocks(self) -> builtins.dict[builtins.str,
builtins.list[builtins.str]]:
```

get_default_runtime_settings

Kind: Method

```
def get_default_runtime_settings(self) -> SettingsValue:
```

get_dot_files

Kind: Method

```
def get_dot_files(self, process: typing.Optional[builtins.int | builtins.str] = None,
integrand_name: typing.Optional[builtins.str] = None, settings: DotExportSettings
= ...) -> builtins.str:
```

process

Kind: Parameter

process: typing.Optional[builtins.int | builtins.str] = None

Default: None

integrand_name

Kind: Parameter

integrand_name: typing.Optional[builtins.str] = None

Default: None

settings

Kind: Parameter

settings: DotExportSettings = ...

Default: ...

run

Kind: Method

def run(self, command: builtins.str) -> None:

command

Kind: Parameter

command: builtins.str

generate_cff

Kind: Method

def generate_cff(self, dot_string: builtins.str, subgraph_nodes: typing.Sequence[builtins.str], reverse_dangling: typing.Sequence[builtins.int], orientation_pattern: typing.Optional[builtins.str] = None) -> builtins.list[tuple[builtins.dict[builtins.int, builtins.int], builtins.str]]:

dot_string

Kind: Parameter

dot_string: builtins.str

subgraph_nodes

Kind: Parameter

subgraph_nodes: typing.Sequence[builtins.str]

reverse_dangling

Kind: Parameter

reverse_dangling: typing.Sequence[builtins.int]

orientation_pattern

Kind: Parameter

orientation_pattern: typing.Optional[builtins.str] = None

Default: None

generate_cff_as_json_string

Kind: Method

def generate_cff_as_json_string(self, dot_string: builtins.str, subgraph_nodes: typing.Sequence[builtins.str], reverse_dangling: typing.Sequence[builtins.int], orientation_pattern: typing.Optional[builtins.str] = None) -> builtins.str:

dot_string

Kind: Parameter

dot_string: builtins.str

subgraph_nodes

Kind: Parameter

subgraph_nodes: typing.Sequence[builtins.str]

reverse_dangling

Kind: Parameter

reverse_dangling: typing.Sequence[builtins.int]

orientation_pattern

Kind: Parameter

orientation_pattern: typing.Optional[builtins.str] = None

Default: None

[Exhaustive reference](#) · [Source](#)

atom_to_canonical_string

Parse a Symbolica expression and return its canonical string form.

Parse a Symbolica expression and return its canonical string form.

```
def atom_to_canonical_string(atom_str: builtins.str) -> builtins.str:
```

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```
from gammaloop._gammaloop import atom_to_canonical_string
help(atom_to_canonical_string)
```

[Exhaustive reference](#) · [Source](#)

evaluate_graph_overall_factor

Evaluate a graph's symbolic overall factor and return its canonical form.

Evaluate a graph's symbolic overall factor and return its canonical form.

```
def evaluate_graph_overall_factor(overall_factor: builtins.str) -> builtins.str:
```

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```
from gammaloop._gammaloop import evaluate_graph_overall_factor
help(evaluate_graph_overall_factor)
```

[Exhaustive reference](#) · [Source](#)

to_dots

Rewrite a Symbolica tensor expression into Idenso dot-product notation.

Rewrite a Symbolica tensor expression into Idenso dot-product notation.

```
def to_dots(atom_str: builtins.str) -> builtins.str:
```

Requires features: python_api, python_abi, python_stubgen

Inspect the registered export

```
from gammaloop._gammaloop import to_dots
help(to_dots)
```

Exhaustive reference · Source

CLI commands and settings

Commands and settings available in this version are listed below.

Command tree

Command	Description
gammaLoop	An implementation of the Local Unitarity method for computing differential collider cross-sections.
gammaLoop display	
gammaLoop display model	
gammaLoop display processes	
gammaLoop display integrand	
gammaLoop display quantities	
gammaLoop display observables	
gammaLoop display selectors	
gammaLoop display command_block	
gammaLoop display settings	
gammaLoop display settings global	
gammaLoop display settings default-runtime	
gammaLoop display settings process	
gammaLoop display settings help	Print this message or the help of the given subcommand(s)
gammaLoop display settings help global	
gammaLoop display settings help default-runtime	
gammaLoop display settings help process	
gammaLoop display settings help help	Print this message or the help of the given subcommand(s)
gammaLoop display help	Print this message or the help of the given subcommand(s)
gammaLoop display help model	
gammaLoop display help processes	
gammaLoop display help integrand	
gammaLoop display help quantities	
gammaLoop display help observables	
gammaLoop display help selectors	
gammaLoop display help command_block	
gammaLoop display help settings	

Command	Description
gammaLoop display help settings global	
gammaLoop display help settings default-runtime	
gammaLoop display help settings process	
gammaLoop display help help	Print this message or the help of the given subcommand(s)
gammaLoop duplicate	
gammaLoop duplicate integrand	
gammaLoop duplicate help	Print this message or the help of the given subcommand(s)
gammaLoop duplicate help integrand	
gammaLoop duplicate help help	Print this message or the help of the given subcommand(s)
gammaLoop set	
gammaLoop set base-dir	Set the base directory for gammaloop state and logs
gammaLoop set global	Set GLOBAL settings
gammaLoop set global file	Load from a settings file
gammaLoop set global string	Load settings from TOML content passed as a CLI string
gammaLoop set global kv	Set one or more dotted key-paths
gammaLoop set global defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set global stored	Use the stored settings file
gammaLoop set global help	Print this message or the help of the given subcommand(s)
gammaLoop set global help file	Load from a settings file
gammaLoop set global help string	Load settings from TOML content passed as a CLI string
gammaLoop set global help kv	Set one or more dotted key-paths
gammaLoop set global help defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set global help stored	Use the stored settings file
gammaLoop set global help help	Print this message or the help of the given subcommand(s)
gammaLoop set default-runtime	Set DEFAULT RUNTIME settings
gammaLoop set default-runtime file	Load from a settings file
gammaLoop set default-runtime string	Load settings from TOML content passed as a CLI string
gammaLoop set default-runtime kv	Set one or more dotted key-paths

Command	Description
gammaLoop set default-runtime defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set default-runtime stored	Use the stored settings file
gammaLoop set default-runtime help	Print this message or the help of the given subcommand(s)
gammaLoop set default-runtime help file	Load from a settings file
gammaLoop set default-runtime help string	Load settings from TOML content passed as a CLI string
gammaLoop set default-runtime help kv	Set one or more dotted key-paths
gammaLoop set default-runtime help defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set default-runtime help stored	Use the stored settings file
gammaLoop set default-runtime help help	Print this message or the help of the given subcommand(s)
gammaLoop set model	Set Model parameters
gammaLoop set process	Set settings for a PROCESS
gammaLoop set process file	Load from a settings file
gammaLoop set process string	Load settings from TOML content passed as a CLI string
gammaLoop set process kv	Set one or more dotted key-paths
gammaLoop set process defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set process stored	Use the stored settings file
gammaLoop set process add	Add a named quantity/observable/selector
gammaLoop set process add quantity	
gammaLoop set process add observable	
gammaLoop set process add selector	
gammaLoop set process add help	Print this message or the help of the given subcommand(s)
gammaLoop set process add help quantity	
gammaLoop set process add help observable	
gammaLoop set process add help selector	
gammaLoop set process add help help	Print this message or the help of the given subcommand(s)
gammaLoop set process update	Update a named quantity/observable/selector

Command	Description
gammaLoop set process update quantity	
gammaLoop set process update observable	
gammaLoop set process update selector	
gammaLoop set process update help	Print this message or the help of the given subcommand(s)
gammaLoop set process update help quantity	
gammaLoop set process update help observable	
gammaLoop set process update help selector	
gammaLoop set process update help help	Print this message or the help of the given subcommand(s)
gammaLoop set process remove	Remove a named quantity/observable/selector
gammaLoop set process remove quantity	
gammaLoop set process remove observable	
gammaLoop set process remove selector	
gammaLoop set process remove help	Print this message or the help of the given subcommand(s)
gammaLoop set process remove help quantity	
gammaLoop set process remove help observable	
gammaLoop set process remove help selector	
gammaLoop set process remove help help	Print this message or the help of the given subcommand(s)
gammaLoop set process help	Print this message or the help of the given subcommand(s)
gammaLoop set process help file	Load from a settings file
gammaLoop set process help string	Load settings from TOML content passed as a CLI string
gammaLoop set process help kv	Set one or more dotted key-paths
gammaLoop set process help defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set process help stored	Use the stored settings file
gammaLoop set process help add	Add a named quantity/observable/selector
gammaLoop set process help add quantity	

Command	Description
gammaLoop set process help add observable	
gammaLoop set process help add selector	
gammaLoop set process help update	Update a named quantity/observable/selector
gammaLoop set process help update quantity	
gammaLoop set process help update observable	
gammaLoop set process help update selector	
gammaLoop set process help remove	Remove a named quantity/observable/selector
gammaLoop set process help remove quantity	
gammaLoop set process help remove observable	
gammaLoop set process help remove selector	
gammaLoop set process help help	Print this message or the help of the given subcommand(s)
gammaLoop set help	Print this message or the help of the given subcommand(s)
gammaLoop set help base-dir	Set the base directory for gammaloop state and logs
gammaLoop set help global	Set GLOBAL settings
gammaLoop set help global file	Load from a settings file
gammaLoop set help global string	Load settings from TOML content passed as a CLI string
gammaLoop set help global kv	Set one or more dotted key-paths
gammaLoop set help global defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set help global stored	Use the stored settings file
gammaLoop set help default-runtime	Set DEFAULT RUNTIME settings
gammaLoop set help default-runtime file	Load from a settings file
gammaLoop set help default-runtime string	Load settings from TOML content passed as a CLI string
gammaLoop set help default-runtime kv	Set one or more dotted key-paths
gammaLoop set help default-runtime defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set help default-runtime stored	Use the stored settings file

Command	Description
gammaLoop set help model	Set Model parameters
gammaLoop set help process	Set settings for a PROCESS
gammaLoop set help process file	Load from a settings file
gammaLoop set help process string	Load settings from TOML content passed as a CLI string
gammaLoop set help process kv	Set one or more dotted key-paths
gammaLoop set help process defaults	Sync process runtime settings from the current default runtime settings
gammaLoop set help process stored	Use the stored settings file
gammaLoop set help process add	Add a named quantity/observable/selector
gammaLoop set help process add quantity	
gammaLoop set help process add observable	
gammaLoop set help process add selector	
gammaLoop set help process update	Update a named quantity/observable/selector
gammaLoop set help process update quantity	
gammaLoop set help process update observable	
gammaLoop set help process update selector	
gammaLoop set help process remove	Remove a named quantity/observable/selector
gammaLoop set help process remove quantity	
gammaLoop set help process remove observable	
gammaLoop set help process remove selector	
gammaLoop set help help	Print this message or the help of the given subcommand(s)
gammaLoop import	
gammaLoop import model	Generate integrands
gammaLoop import graphs	
gammaLoop import help	Print this message or the help of the given subcommand(s)
gammaLoop import help model	Generate integrands
gammaLoop import help graphs	
gammaLoop import help help	Print this message or the help of the given subcommand(s)
gammaLoop save	

Command	Description
gammaLoop save dot	
gammaLoop save uv-forest	
gammaLoop save standalone	
gammaLoop save state	
gammaLoop save schema	regenerate the schema files
gammaLoop save help	Print this message or the help of the given subcommand(s)
gammaLoop save help dot	
gammaLoop save help uv-forest	
gammaLoop save help standalone	
gammaLoop save help state	
gammaLoop save help schema	regenerate the schema files
gammaLoop save help help	Print this message or the help of the given subcommand(s)
gammaLoop run	
gammaLoop start_commands_block	
gammaLoop finish_commands_block	
gammaLoop remove	
gammaLoop remove processes	
gammaLoop remove help	Print this message or the help of the given subcommand(s)
gammaLoop remove help processes	
gammaLoop remove help help	Print this message or the help of the given subcommand(s)
gammaLoop integrate	
gammaLoop generate	Generate integrands
gammaLoop generate xs	Cross-section (forward-scattering) generation
gammaLoop generate amp	Amplitude generation
gammaLoop generate existing	Reuse an already generated process
gammaLoop generate help	Print this message or the help of the given subcommand(s)
gammaLoop generate help xs	Cross-section (forward-scattering) generation
gammaLoop generate help amp	Amplitude generation
gammaLoop generate help existing	Reuse an already generated process
gammaLoop generate help help	Print this message or the help of the given subcommand(s)
gammaLoop select	
gammaLoop quit	Quit gammaloop
gammaLoop inspect	Inspect a single phase-space point / momentum configuration
gammaLoop approach	Approach a phase-space point along one or more axes

Command	Description
gammaLoop evaluate	
gammaLoop renormalize	
gammaLoop bench	Benchmark raw integrand evaluation speed
gammaLoop profile	
gammaLoop profile ultra-violet	Ultraviolet profile analysis
gammaLoop profile bulk	Bulk profile analysis
gammaLoop profile help	Print this message or the help of the given subcommand(s)
gammaLoop profile help ultra-violet	Ultraviolet profile analysis
gammaLoop profile help bulk	Bulk profile analysis
gammaLoop profile help help	Print this message or the help of the given subcommand(s)
gammaLoop batch	HPC batch evaluation branch
gammaLoop !	
gammaLoop help	Print this message or the help of the given subcommand(s)
gammaLoop help display	
gammaLoop help display model	
gammaLoop help display processes	
gammaLoop help display integrand	
gammaLoop help display quantities	
gammaLoop help display observables	
gammaLoop help display selectors	
gammaLoop help display command_block	
gammaLoop help display settings	
gammaLoop help display settings global	
gammaLoop help display settings default-runtime	
gammaLoop help display settings process	
gammaLoop help duplicate	
gammaLoop help duplicate integrand	
gammaLoop help set	
gammaLoop help set base-dir	Set the base directory for gammaloop state and logs
gammaLoop help set global	Set GLOBAL settings
gammaLoop help set global file	Load from a settings file

Command	Description
gammaLoop help set global string	Load settings from TOML content passed as a CLI string
gammaLoop help set global kv	Set one or more dotted key-paths
gammaLoop help set global defaults	Sync process runtime settings from the current default runtime settings
gammaLoop help set global stored	Use the stored settings file
gammaLoop help set default-runtime	Set DEFAULT RUNTIME settings
gammaLoop help set default-runtime file	Load from a settings file
gammaLoop help set default-runtime string	Load settings from TOML content passed as a CLI string
gammaLoop help set default-runtime kv	Set one or more dotted key-paths
gammaLoop help set default-runtime defaults	Sync process runtime settings from the current default runtime settings
gammaLoop help set default-runtime stored	Use the stored settings file
gammaLoop help set model	Set Model parameters
gammaLoop help set process	Set settings for a PROCESS
gammaLoop help set process file	Load from a settings file
gammaLoop help set process string	Load settings from TOML content passed as a CLI string
gammaLoop help set process kv	Set one or more dotted key-paths
gammaLoop help set process defaults	Sync process runtime settings from the current default runtime settings
gammaLoop help set process stored	Use the stored settings file
gammaLoop help set process add	Add a named quantity/observable/selector
gammaLoop help set process add quantity	
gammaLoop help set process add observable	
gammaLoop help set process add selector	
gammaLoop help set process update	Update a named quantity/observable/selector
gammaLoop help set process update quantity	
gammaLoop help set process update observable	
gammaLoop help set process update selector	
gammaLoop help set process remove	Remove a named quantity/observable/selector

Command	Description
gammaLoop help set process remove quantity	
gammaLoop help set process remove observable	
gammaLoop help set process remove selector	
gammaLoop help import	
gammaLoop help import model	Generate integrands
gammaLoop help import graphs	
gammaLoop help save	
gammaLoop help save dot	
gammaLoop help save uv-forest	
gammaLoop help save standalone	
gammaLoop help save state	
gammaLoop help save schema	regenerate the schema files
gammaLoop help run	
gammaLoop help start_commands_block	
gammaLoop help finish_commands_block	
gammaLoop help remove	
gammaLoop help remove processes	
gammaLoop help integrate	
gammaLoop help generate	Generate integrands
gammaLoop help generate xs	Cross-section (forward-scattering) generation
gammaLoop help generate amp	Amplitude generation
gammaLoop help generate existing	Reuse an already generated process
gammaLoop help select	
gammaLoop help quit	Quit gammaloop
gammaLoop help inspect	Inspect a single phase-space point / momentum configuration
gammaLoop help approach	Approach a phase-space point along one or more axes
gammaLoop help evaluate	
gammaLoop help renormalize	
gammaLoop help bench	Benchmark raw integrand evaluation speed
gammaLoop help profile	
gammaLoop help profile ultra-violet	Ultraviolet profile analysis
gammaLoop help profile bulk	Bulk profile analysis
gammaLoop help batch	HPC batch evaluation branch
gammaLoop help !	

Path	Type	Default
cli.global.generation.tropical_subgraphable.panic_on_fail	boolean	false
cli.global.generation.tropical_subgraphable.target_omega	number	1.0
cli.global.generation.uv.add_markers	boolean	false
cli.global.generation.uv.final_integrating	string	"ThreeD"
cli.global.generation.uv.generate_bonded	boolean	true
cli.global.generation.uv.inner_products	boolean	true
cli.global.generation.uv.keep_markers	boolean	true
cli.global.generation.uv.orchestrating	string	"hedge_poset"
cli.global.generation.uv.renormalization_prescription_MUg_divergent	string	"MUg_divergent"
cli.global.generation.uv.renormalization_prescription_MUgless_power_divergent	string	"MUgless_power_divergent"
cli.global.generation.uv.renormalization_prescription_MUglessless_power_divergent	string	"MUglessless_power_divergent"
cli.global.generation.uv.renormalization_prescription Overrides	string	" Overrides"
cli.global.generation.uv.softcut	boolean	true
cli.global.generation.uv.subtract_bonds	boolean	true
cli.global.generation.uv.vakint.additional_normalization	boolean	true
cli.global.generation.uv.vakint.alphaloop_substitution_masters	boolean	true
cli.global.generation.uv.vakint.cleanup_dir	boolean	true
cli.global.generation.uv.vakint.evaluation_methods	array	["alphaloop", "matad", "fmft"]
cli.global.generation.uv.vakint.expand_masters	boolean	true
cli.global.generation.uv.vakint.expand_substitution_masters	boolean	true
cli.global.generation.uv.vakint.forming_path	string	"form"
cli.global.generation.uv.vakint.match_direct_numerical_substitution	boolean	true
cli.global.generation.uv.vakint.match_expand_master	boolean	true
cli.global.generation.uv.vakint.match_expand_substitution_hosts	boolean	true
cli.global.generation.uv.vakint.match_expand_substitution_masters	boolean	true
cli.global.generation.uv.vakint.normalization	string	"MSbar"
cli.global.generation.uv.vakint.pysetgemax_n_eval	integer	1000000000000
cli.global.generation.uv.vakint.pysetgemin_n_eval	integer	10000
cli.global.generation.uv.vakint.pysetquiet	boolean	true
cli.global.generation.uv.vakint.pysetnumber.relative_precision	number	1e-7
cli.global.generation.uv.vakint.pysetlog.reuse_existing_output	boolean	reused default
cli.global.generation.uv.vakint.pythontmp_exe_path	string	"python3"
cli.global.generation.uv.vakint.runtimedecimal_precision	integer	100
cli.global.generation.uv.vakint.temporary_directory	string	no serialized default
cli.global.generation.vector_polarization_sum_gauge	string	"LightLikeAxial"
cli.global.log_style.full_line_source	boolean	false
cli.global.log_style.include_fields	boolean	false
cli.global.log_style.log_format	string	"Long"
cli.global.log_style.short_timestamp	boolean	false
cli.global.logfile_directive	string	"off"

Path	Type	Default
cli.global.n_cores.compile	integer	1
cli.global.n_cores.feyngen	integer	1
cli.global.n_cores.generate	integer	1
cli.global.n_cores.integrate	integer	1
cli.override_state	boolean	false
cli.state.folder	string	"./gamma_loop_state"
cli.state.name	string null	""
cli.try_strings	boolean	true
runtime.general.additional_param_values	array	[]
runtime.general.debug_cache	boolean	false
runtime.general.disable_flux_factor	boolean	false
runtime.general.enable_cache	boolean	false
runtime.general.evaluator_method	string	"SingleParametric"
runtime.general.generate_events	boolean	false
runtime.general.integral_unit	string	"auto"
runtime.general.load_compiled_cff	boolean	false
runtime.general.m_uv	number	1000.0
runtime.general.mu_r	number	1000.0
runtime.general.orientation_path	null string	no serialized default
runtime.general.renormalization_localization_scale	number	1000.0
runtime.general.store_additional_weights_in_event	boolean	false
runtime.general.use_ltd	boolean	false
runtime.h_function.enabled_dampening	boolean	true
runtime.h_function.function	string	"poly_exponential"
runtime.h_function.power	integer null	no serialized default
runtime.h_function.sigma	number	1.0
runtime.integrand.n_3d_momenta	integer	no serialized default
runtime.integrator.bin_number_evolutionary	array null	no serialized default
runtime.integrator.continuous_dim_learning_rate	number	1.0
runtime.integrator.discrete_dim_learning_rate	number	1.0
runtime.integrator.integrated_phase	string	"real"
runtime.integrator.max_prob_ratio	number	30.0
runtime.integrator.min_samples_for_integration	integer	1000
runtime.integrator.n_bins	integer	64
runtime.integrator.n_increase	integer	10000
runtime.integrator.n_max	integer	100000000000
runtime.integrator.n_start	integer	100000
runtime.integrator.observables_output_format	string	["json"]
runtime.integrator.seed	integer	69
runtime.integrator.show_max_wgt_info	boolean	true

Path	Type	Default
runtime.integrator.target_absolute_number	number	no serialized default
runtime.integrator.target_relative_number	number	no serialized default
runtime.integrator.train_on_avg	boolean	false
runtime.kinematics.e_cm	number	64.0
runtime.kinematics.externals.data.activities	array	[]
runtime.kinematics.externals.data.initial_member_settings	array	large_deformation_check
runtime.kinematics.externals.data.improvement_settings	array	none
runtime.kinematics.externals.data.improvement_settings	array	only_warn_on_large_deformation
runtime.kinematics.externals.data.moments	array	[]
runtime.kinematics.externals.type	string	"constant"
runtime.model	object	{}
runtime.observables	object	{}
runtime.quantities	object	{}
runtime.sampling.alpha	number	3.0
runtime.sampling.b	number	1.0
runtime.sampling.coordinate_system	string	"spherical"
runtime.sampling.graph_names	array	[]
runtime.sampling.graphs	string	"summed"
runtime.sampling.lmb_basis_ids	object	no serialized default
runtime.sampling.lmb_channel_weight	string	"ose"
runtime.sampling.lmb_channels	string	"summed"
runtime.sampling.lmb_multichanneling	boolean	false
runtime.sampling.mapping	string	"linear"
runtime.sampling.orientations	string	"summed"
runtime.sampling.power	number	1.0
runtime.selectors	object	{}
runtime.stability.check_on_norm	boolean	true
runtime.stability.escalate_if_exact	boolean	false
runtime.stability.levels	array	[{"escalate_for_large_weight_threshold":0.9,"pr {"escalate_for_large_weight_threshold":-1.0,"pr {"escalate_for_large_weight_threshold":-1.0,"pr
runtime.stability.loop_momenta_norm	number	1.0
runtime.stability.recording.recordable_stability_level	boolean	no serialized default
runtime.stability.recording.recordable_momenta_escalation	boolean	no serialized default
runtime.stability.recording.recordable_results	boolean	no serialized default
runtime.stability.rotation_axis	array	[{"alpha":0.1,"beta":0.2,"gamma":0.3,"type":"eu
runtime.subtraction.disable_threshold_subtraction	boolean	false
runtime.subtraction.esurface_existence_threshold	number	1e-7
runtime.subtraction.integrated_ct_settings.range.h	function	function_settings.enabled_dampening
runtime.subtraction.integrated_ct_settings.range.h	function	function_settings.function
runtime.subtraction.integrated_ct_settings.range.h	function	function_settings.power

Path	Type	Default
runtime.subtraction.integrated_ct_settings.range.h	float	function_settings.sigma
runtime.subtraction.integrated_ct_settings.range.type	string	"infinite"
runtime.subtraction.local_ct_settings.localisation.dynamic_width	boolean	false
runtime.subtraction.local_ct_settings.localisation.force_uv_dampers_to_one	boolean	false
runtime.subtraction.local_ct_settings.localisation.gaussian_width	number	0.0
runtime.subtraction.local_ct_settings.localisation.oliver_width	number	0.0
runtime.subtraction.overlap_settings.back_global_center	boolean	true
runtime.subtraction.overlap_settings.by_default_global_center	boolean	serialized default
runtime.subtraction.overlap_settings.local_origin	boolean	true
runtime.subtraction.overlap_settings.local_origin_all	boolean	false
runtime.subtraction.radial_root_residual_tolerance	number	64.0

Version information

Save these identifiers with published results and include them in issue reports so that collaborators can reproduce the same software environment.

- **Documentation channel:** latest
- **Documentation route:** /products/gammaploop/latest/
- **Source revision:** f46892d216f1dafa21e0fbbebd2c69c328f47fa2

Package versions and enabled features

- gammaploop/gammaplooprs — 0.3.4 (no optional features)
- gammaploop/gammaploop-api — 0.1.0 (features: cli)
- gammaploop/gammaploop — 0.3.4 (features: python_api, python_abi, python_stubgen)
- linnet/linnet — 0.17.0 (features: nodestore-vec, serde, bincode, drawing, symbolica)
- linnet/linnet-py — 0.1.0 (features: extension-module, abi3-py310)
- spenso/spenso — 0.6.0 (features: shadowing)
- spenso/spenso-macros — 0.3.2 (features: shadowing)
- spenso/spenso-hep-lib — 0.1.7 (no optional features)
- spenso/spynso3 — 0.1.2 (features: python_stubgen)
- idenso/idenso — 0.3.0 (features: bincode, reference-cases)
- idenso/idenso — 0.3.0 (features: python, python_stubgen)
- vakint/vakint — 0.1.2 (no optional features)
- vakint/vakint — 0.1.2 (features: symbolica_community_module, python_stubgen)